

Data Analytics Insights Report

Metro Interstate Traffic Volume — Westbound I-94, Minneapolis–St Paul

Key Findings

- **Coverage gaps distort raw yearly totals.** Sensor coverage ranged from 24% of hours (2012) to 99.5% (2017); 2012, 2014 and 2015 are each missing several consecutive months. Average hourly volume — not the raw yearly total — is the only fair year-over-year comparison.
- **Traffic is trending up once coverage is normalized.** Average hourly volume dipped through 2014–2016 (3,270 → 3,258 → 3,194 veh/hr) then rose to 3,377 in 2017, its highest of the six years (+5.7% vs. the 2016 low), in the two most completely-measured years.
- **Holidays reliably suppress traffic, independent of weather.** New Year’s Day and Labor Day traffic both sit far below the 3,291 veh/hr baseline every year measured, even as their weather varies year to year (e.g. Labor Day cooled 22.2°C→18.1°C from 2015–2017 while its traffic did not move in step).
- **Temperature is a weak, non-linear predictor of volume.** Pearson $r = 0.137$ (temp vs. volume) — temperature alone explains under 2% of the variance. Volume is fairly flat from –20°C to 20°C, then rises sharply above 25°C (avg. 4,419–4,785 veh/hr) and drops below –25°C (avg. 2,081 veh/hr).
- **Congestion (>5,500 veh/hr) is measurably less likely in clear weather.** $P(\text{Congestion})=15.1\%$, $P(\text{Clear})=32.9\%$. The two are not independent (observed joint probability 4.34% vs. 4.96% expected): the odds of congestion in clear weather are about 26% lower than in cloudy weather (odds ratio 0.74) — likely confounded by time-of-day/season rather than a direct weather effect.
- **Weather-condition impact is real but modest.** Among conditions with a meaningful sample, “Clouds” carries the highest average volume (3,617 veh/hr) and “Fog” the lowest (2,666), a difference of about 951 veh/hr — clearer skies do not mean heavier traffic on this corridor.
- **The raw data contains sensor errors that would corrupt downstream modeling if left untreated.** One hour records 9,831 mm of rainfall (physically impossible), 10 hours record 0 Kelvin, and 2 daytime hours record 0 traffic volume with no holiday/incident explanation.

Implications for the Mobility Team

- **Report and plan on rate metrics, not raw totals.** Any capacity or budget comparison across 2012–2016 must use average hourly volume; raw yearly totals from those years will materially understate true demand.
 - **Schedule lane closures and maintenance around holidays.** Both federal holidays tested show reliably suppressed demand regardless of weather — low-risk windows for planned works.
 - **Don’t rely on weather alone to trigger traffic-response protocols.** Both the temperature correlation and the weather/congestion odds ratio are too weak to drive automated signal-timing or incident-response changes by themselves; combine with time-of-day and day-of-week signals (the natural next step for Part 2/3 feature engineering).
 - **Watch the warm-weather tail.** The highest observed average volumes occur above 25°C — rare in this region but worth targeted monitoring capacity during heat events and the summer construction/event season.
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